
```

clc; clear; close all;

% Example function (Same as lecture and AI)
f = @(x) x.^3 - 6*x.^2 + 11*x - 6;

% Plot function
x = linspace(0,4,100);
y = f(x);

figure
plot(x,y,'-', 'LineWidth',2)
grid on, hold on

xlabel('x')
ylabel('f(x)')
title('Incremental Search for Root Brackets')

% Add line f(x)=0
plot([0 4],[0 0], 'r--')

% As used in the lecture
xb = incsearch(f,0,4,100); % I just rewrote the script and used the
incsearch provided

% Displaying the brackets
disp('Bracket Intervals:')
disp(xb)

% Plot brackets on graph
[m,n] = size(xb);

for i = 1:m
    xL = xb(i,1);
    xU = xb(i,2);

    % Plotting bracket endpoints
    plot(xL,f(xL), 'ko', 'MarkerFaceColor','g', 'MarkerSize',10)
    plot(xU,f(xU), 'ko', 'MarkerFaceColor','m', 'MarkerSize',10)

    % xline for the brackets
    xline(xL, 'b--', 'LineWidth',2)
    xline(xU, 'm--', 'LineWidth',2)

    fprintf('Bracket %d: [%.4f , %.4f]\n',i,xL,xU);
end

Number of Brackets:
    3

Bracket Intervals:
    0.9697    1.0101
    1.9798    2.0202

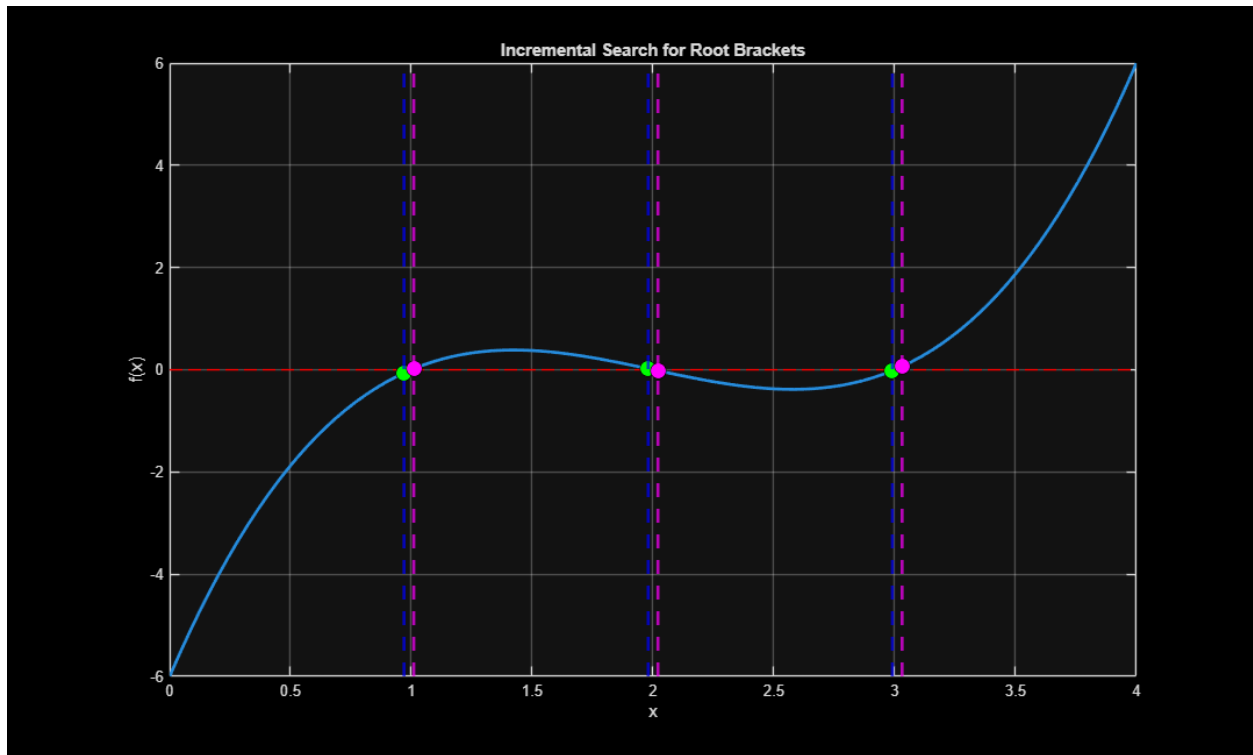
```

2.9899 3.0303

Bracket 1: [0.9697 , 1.0101]

Bracket 2: [1.9798 , 2.0202]

Bracket 3: [2.9899 , 3.0303]



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